

Saurav Chowdhury

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LinkedIn — GitHub — Medium — Google Scholar

Professional Summary

AI/ML Engineer with 8+ years of experience in Natural Language Processing (NLP) and large language models (LLMs), with applied work in legal AI, cybersecurity, and domain-specific language systems. Experience includes language model fine-tuning, semantic analysis, and development of retrieval-based systems, complemented by project-level coordination of AI initiatives. Interested in advancing domain-adaptive and interpretable NLP systems for real-world impact. Actively involved in teaching and mentorship, with a focus on developing strong mathematical foundations and supporting learners in machine learning and natural language processing.

Skills

- **Languages & Frameworks:** Python, PyTorch, TensorFlow, Scikit-learn
- **NLP & LLMs:** GPT, LLaMA, BERT, RAG, LoRA, Quantization, Fine-tuning, Prompt Engineering, NER
- **Deep Learning:** LSTM, CNN, Time-Series Forecasting
- **Tools:** Docker, Hugging Face, Neo4j, WandB, Pandas
- **Domains:** Legal Tech, Enterprise AI Systems, Cybersecurity

Work Experience

Deputy Manager – IT

Assam Power Distribution Company Ltd.

Jan 2022 – Present

- Fine-tuned LLaMA-based models for legal document summarization using QLoRA, integrated within a Retrieval-Augmented Generation (RAG) framework to enable efficient and context-aware information retrieval
- Automated SAP-ERP purchase order amendment workflows, reducing processing time by approximately 90%
- Contributed to ISMS compliance initiatives aligned with CERT-IN and NCIIPC guidelines, supporting secure AI system deployment in critical infrastructure settings
- Core contributor to AI/ML-driven Business Intelligence systems across 12+ modules including forecasting, loss reduction, and event analytics, with involvement in data preprocessing, model fine-tuning, and deployment
- Developed NLP-driven analytics for CRM data, enabling extraction of consumer insights through text classification and pattern analysis to support operational decision-making
- Worked on data-driven electrical network indexing using time-series voltage analytics and correlation-based validation on smart meter (AMI) data.
- Developed an offline LLM-based intelligent query routing system for cybersecurity documents and policy corpora, enabling efficient semantic search and context-aware information retrieval in secure environments

Assistant Manager – IT

Assam Power Distribution Company Ltd.

Jan 2018 – Jan 2022

- Worked closely with the Managing Director's office, gaining exposure to end-to-end business operations, monitoring frameworks, and decision-making workflows
- Leveraged this system-level understanding to identify opportunities for data-driven optimization and AI-based interventions
- Developed machine learning models for large-scale infrastructure and energy datasets
- Built forecasting pipelines supporting renewable energy planning and analytics

Risk Analyst

Amazon Development Centre (India) Pvt. Ltd.

Jan 2017 – Aug 2017

- Analyzed large-scale transactional data to identify fraud patterns and anomalous user behavior, and developed an automated annotation tool to streamline investigation workflows and improve labeling efficiency

Independent Research

- Developing LLM-based legal intelligence systems, including mapping between IPC, CrPC and BNS/BNSS for structured legal knowledge bases
- Building retrieval and embedding pipelines for domain-specific corpora including ancient Indian texts (Ax-omsastra initiative)
- Working on multimodal and domain-adaptive LLM systems for legal compliance analysis (POCSO, FSSAI)

Education

M.Tech – Artificial Intelligence

IIT Jodhpur

2022–2024

CGPA: 8.4

- LLM-based semantic analysis of POCSO judgments
- Developed summarization pipeline reducing legal document length by 85%
- Built ontology-driven QA system using RAG (84% accuracy)

B.Tech – Computer Science & Engineering

Jorhat Engineering College

2012–2016

First Class

Publications

- **Cross Examine: Leveraging LLMs for Legal Text Analytics**, NLLP, EMNLP 2024, Miami, USA
Saurav Chowdhury, Suyog Joshi, Lipika Dey
- **Collective Learning-Based Optimal Transport GAN with Multi-Level Fine-Grained and Global Discriminators for Voice Conversion**, 2025 Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)
Sandipan Dhar, Md Tousin Akhter, Nanda Dulal Jana, Swagatam Das, Monorama Swain, Saurav Chowdhury
- **Intelligent Query Routing for Critical Infrastructure Security: An Offline Information Retrieval Architecture (DuoMind)**
Preprint
- **Artificial Intelligence and Youth Perceptions in Northeast India: Implications for Education and Inclusive Development**
Ankita Choudhury, Saurav Chowdhury
Presented at Viksit Bharat Seminar, TISS
Preprint

Certifications

- Machine Learning – Stanford University (Coursera), Prof. Andrew Ng, Jul 2022
- Generative AI – SAP (Online Course), Nov–Dec 2023

Languages

Hindi, Bengali, Assamese, English